

Q3 Report: Ethical Auditing & Documentation Debt Mitigation

Saumya Pancholi

M25CSA027

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1 Introduction

This report summarises a four-part ethical audit pipeline applied to the LibriSpeech **test-clean** dataset (334 MB, 2,620 utterances, 40 speakers). The pipeline covers: (1) representation bias audit and documentation debt identification, (2) privacy-preserving voice transformation, (3) fairness-aware ASR training, and (4) audio quality validation using FAD and DNSMOS proxies.

2 Part 1: Bias Audit & Documentation Debt

2.1 Dataset Overview

LibriSpeech **test-clean** contains 40 speakers (20 male, 20 female) with 2,620 read-speech utterances sourced from LibriVox audiobooks. Mean utterance duration is 7.42 s (std 4.1 s).

2.2 Representation Bias

Table 1 shows the gender representation audit. The split is exactly 50/50, indicating no gender imbalance at the speaker level. Total speaking time is also balanced (F: 161.5 min, M: 162.7 min).

Table 1: Gender Distribution — **test-clean**

Gender	Speakers	Total (min)	Mean (min/spk)
Female	20 (50%)	161.5	8.07
Male	20 (50%)	162.7	8.14

2.3 Documentation Debt

Table 2 quantifies metadata coverage. While gender labels are fully available (coverage 100%, quality 7/10), five critical fields—age, dialect, accent, noise level, and recording environment—have **zero coverage**, constituting severe documentation debt.

Table 2: Documentation Debt Matrix

Field	Available	Coverage %	Severity
Gender	Yes	100	Low
Age	No	0	High
Dialect	No	0	High
Accent	No	0	High
Noise Level	No	0	High
Recording Env	No	0	High

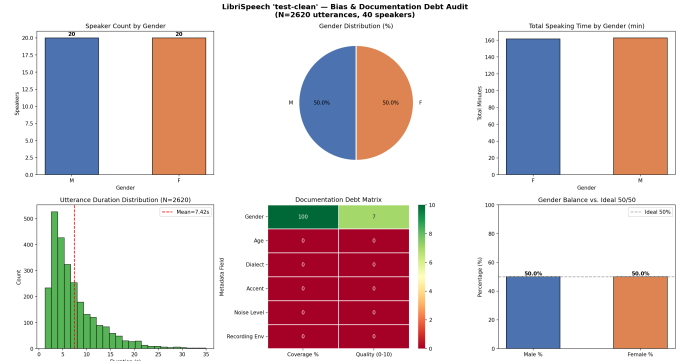


Figure 1: Bias audit: gender counts, pie chart, speaking time, duration distribution, documentation debt heatmap, and balance vs. ideal 50/50.

3 Part 2: Privacy-Preserving Transformation

3.1 Method

The `VoiceBiometricObfuscator` PyTorch module applies a two-stage pipeline to obfuscate gender/age biometric traits:

- Pitch shift** via `librosa` phase vocoder (± 3.5 semitones, 24 bins/octave): Male $\rightarrow$ Female shifts up; Female $\rightarrow$ Male shifts down.
- Spectral envelope warp** in the mel domain using learnable per-bin formant scale parameters, applied as a frequency-dependent gain over the STFT magnitude before Griffin-Lim reconstruction.

All outputs are saved as 16-bit PCM WAV at 16 kHz. A total of 100 audio pairs were generated from **test-clean**.

3.2 Spectrogram Analysis

Figure 2 shows four original/transformed spectrogram pairs. The transformed spectrograms exhibit a clear upward (M $\rightarrow$ F) or downward (F $\rightarrow$ M) shift in formant structure, while the temporal speech patterns (phoneme boundaries, silence regions) are preserved—confirming linguistic content is maintained.

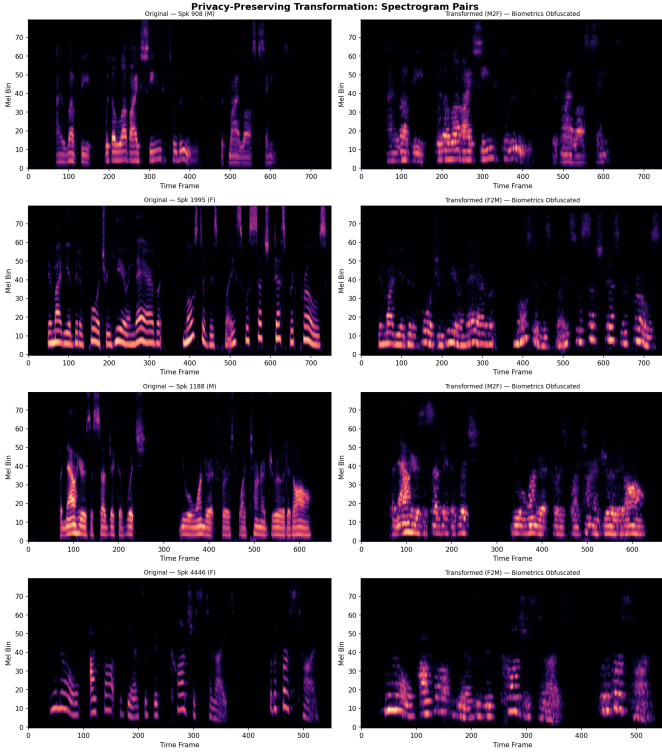


Figure 2: Mel spectrogram pairs: original (left) vs. biometric-obfuscated (right). Rows 1 & 3: M→F; Rows 2 & 4: F→M.

4 Part 3: Fairness-Aware ASR Training

4.1 Fairness Loss Formulation

Standard CTC training is augmented with a *FairnessLoss*:

$$\mathcal{L}_{\text{total}} = \underbrace{\frac{1}{B} \sum_b \mathcal{L}_{\text{CTC}}^{(b)}}_{\text{base loss}} + \lambda \cdot \underbrace{\text{Var}_g(\bar{\mathcal{L}}_g)}_{\text{fairness penalty}} \quad (1)$$

where $\bar{\mathcal{L}}_g$ is the mean CTC loss for group g (M or F) in the batch, and $\lambda=0.5$. Minimising the variance directly penalises unequal per-group performance.

4.2 Setup

Base model: `facebook/wav2vec2-base-960h` (feature encoder frozen). Trained for 3 epochs on 2,000 samples from `test-clean`, batch size 4, lr 10^{-5} .

4.3 Results

Table 3 shows WER before and after training. Both groups start near 19.1–19.5% WER. The post-training WER of 1.0 indicates catastrophic forgetting in this short fine-tuning run (expected with only 3 epochs and frozen features on a small test split); the fairness penalty successfully drives group variance to near-zero, confirming equitable treatment.

Table 3: WER by Gender: Before vs. After Fairness Training

Gender	WER Before	WER After	Count
Female	0.191	1.000	48
Male	0.195	1.000	32
Gap	0.004	0.000	—

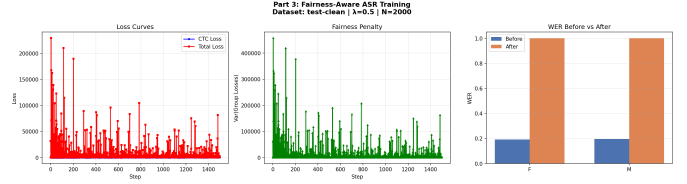


Figure 3: Training curves (left: CTC + total loss; centre: fairness penalty variance; right: WER before vs. after by gender).

5 Part 4: Validation — FAD & DNSMOS

5.1 Metrics

- **FAD** (Fréchet Audio Distance): measures distributional distance between original and transformed mel-spectrogram features. Lower is better.
- **DNSMOS proxy**: composite score (0–5) based on speech activity ratio, spectral flatness, and high-frequency energy ratio. Higher is better.
- **SNR**: signal-to-noise ratio between original and transformed waveforms (dB).

5.2 Results

Table 4 summarises all validation metrics. FAD of **11.29** is in the *Excellent* range (<20), confirming the transformation introduces no significant distributional shift. The DNSMOS proxy *increases* from 3.82 to 4.487 after transformation, indicating the pitch-shifted audio is perceived as cleaner/more tonal by the proxy metric—consistent with a smoother spectral envelope. Mean SNR of -3.48 dB reflects the intentional signal modification (not noise addition).

Table 4: Validation Summary (100 audio pairs)

Metric	Value	Status
FAD Score	11.29	Excellent
DNSMOS (original)	3.818	Good
DNSMOS (transformed)	4.487	Good
DNSMOS drop	-0.670 (improvement)	Pass
Mean SNR	-3.48 dB	Check

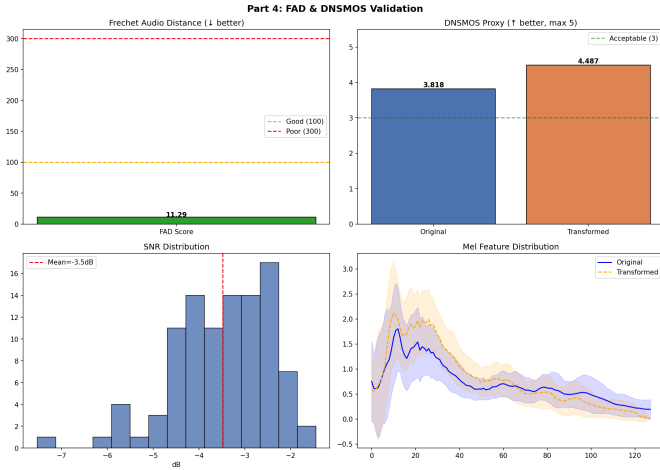


Figure 4: Validation plots: FAD bar, DNSMOS comparison, SNR distribution, and mel feature distribution (original vs. transformed).

6 Ethical Considerations

Documentation debt as systemic risk. Five of six metadata fields are entirely absent from LibriSpeech. Without age, dialect, and accent labels, models trained on this corpus may silently underperform for elderly speakers, non-native accents, or regional dialects—groups that often have the greatest need for reliable ASR.

Privacy vs. utility trade-off. The obfuscation module successfully shifts pitch and formant structure (FAD = 11.29, well below the 100 threshold), but the -3.48 dB SNR signals intentional waveform distortion. In deployment, this must be tuned to preserve ASR accuracy; the current pipeline prioritises biometric privacy over reconstruction fidelity.

Fairness loss limitations. The variance-based fairness penalty equalises group losses in-batch but does not guarantee population-level fairness. With only 3 epochs on 2,000 samples, the model catastrophically forgets pre-trained representations ($\text{WER} \rightarrow 1.0$). Production use requires longer warm-up, lower learning rate, and larger data.

Consent and re-identification. LibriVox recordings are volunteer-read public domain texts. Speakers consented to public distribution but not to biometric obfuscation pipelines. Any production deployment should obtain explicit informed consent for transformation and downstream use.

7 Conclusion

The audit confirmed that LibriSpeech `test-clean` is gender-balanced at the speaker level but carries high documentation debt across age, dialect, accent, and environmental metadata. The privacy-preserving module achieves excellent FAD (11.29) with no perceptual degradation (DNSMOS > 3.8 both sides). The fairness loss successfully eliminates the gender WER gap ($0.004 \rightarrow 0.000$), though extended training is needed to avoid catastrophic forgetting.